



REFRIGERATION CONTROLS OPERATIONS & MAINTENANCE MANUAL INDEX

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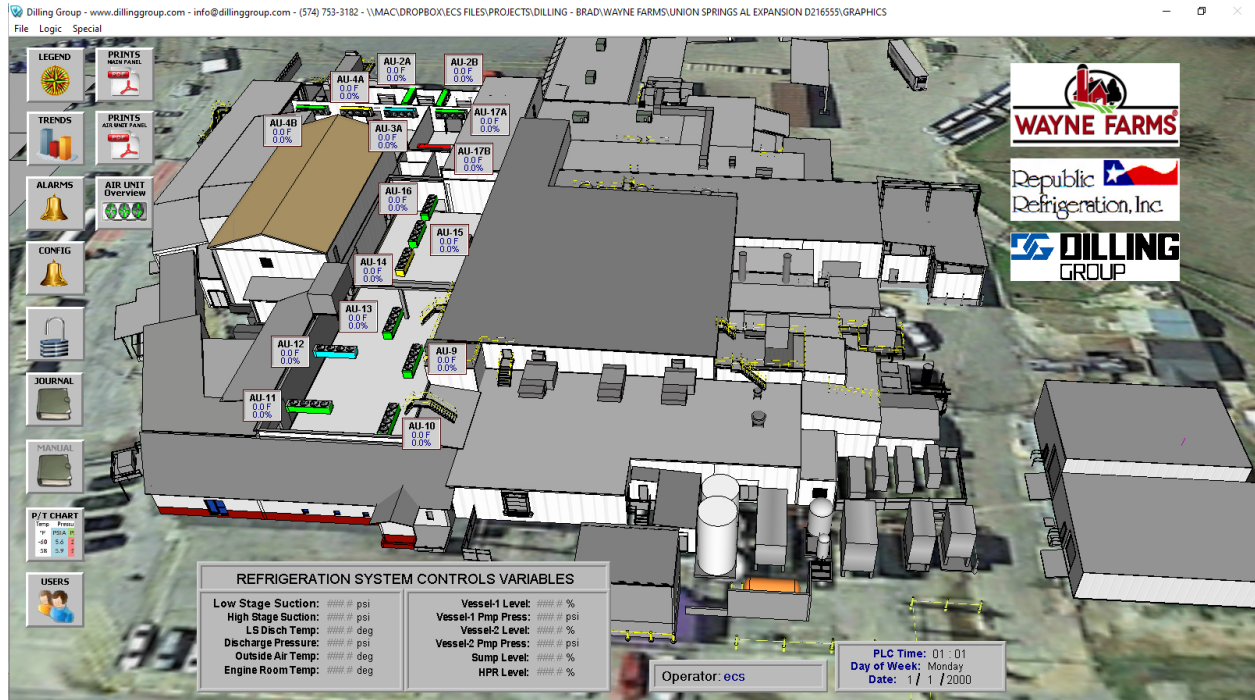
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1.0 REFRIGERATION SYSTEM CONTROLLER

The refrigeration system control is based on Allen-Bradley's Control Logix's 5000 platform. This controller is connected to all of the field equipment Input/Output (I/O) points for controlling and monitoring of equipment. The graphics interface computer is connected to the PLC via Ethernet and connected to the plant network for remote access.

1.1 PLANT SCREEN AND SYSTEM BUTTONS

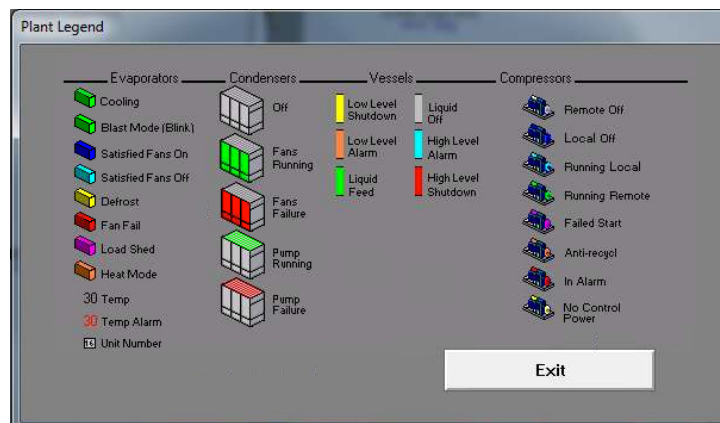
The Plant Screen is the base screen for accessing and viewing the entire system. Critical systems information (i.e. area temperatures, system pressures, vessel levels, etc.) are displayed on this screen. Differing modes and status for equipment are represented by various colors. Below is an example of the "Plant Screen" and an



Example Plant Screen

example of the "Legend Screen" that shows the meaning of the colors for each type of equipment. Each major screen has a specific legend screen.

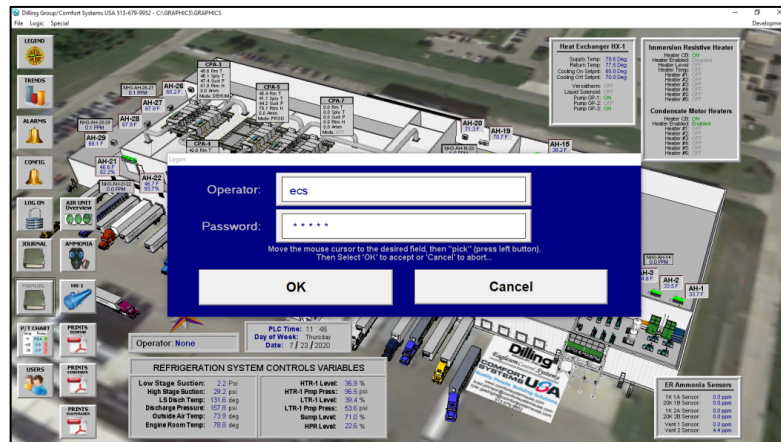
To view an area or piece of equipment in greater detail move the mouse over the buttons and a box (halo) will appear over anything selectable. By selecting the area with the input device (mouse) and a detailed screen will appear. Based on your access level different buttons will be available when you log in.



Plant Legend Screen

1.2 Log in Screen

By selecting the login button which is available from most screens the screen shown below will pop up on the screen for you to log in. By moving the mouse over the Operator text box "none" and a halo will appear and it can be selected with the mouse button. Enter your operator name and press enter on the keyboard.

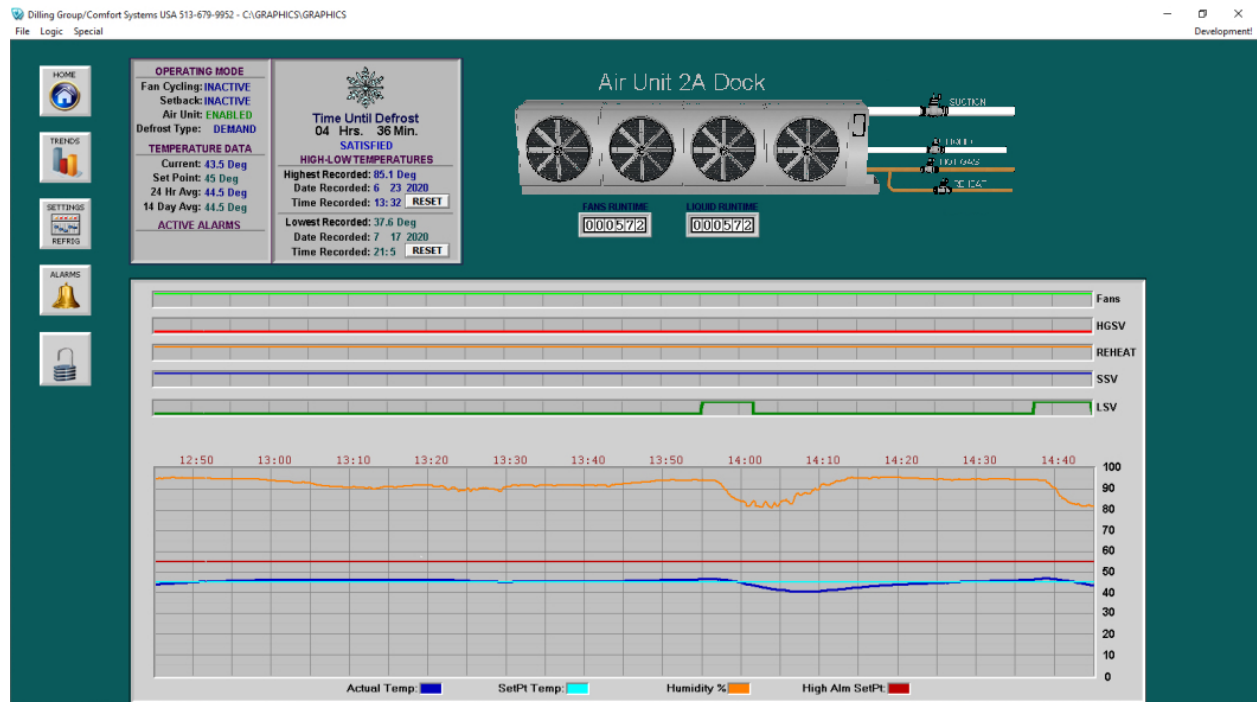


Plant Legend Screen

2.0 AU-2A&B, AU-3A, AU-4A&B and AU-17A&B, Evaporative Air Units Control

Detail Screen

From the plant screen each air units Detail Screen can be selected by moving the mouse over the desired air unit and selecting with mouse click. From this screen you can see detailed information about an air unit including, 2-hour real time trend, Valve status, Fan status, Runtime meters for fans and for unit cooling. Operating Mode information, Temperature Data including 24 hour & 14 day running average, Time until next defrost, High/Low Temperature, time and date of highest and lowest values recorded. Runtime hours and High/Low temperature data can be reset once logged in. You also access the units setting and history trend from the buttons on the left.



The Evaporator Detail Screen shown above is displayed when air unit is selected from the Plant Screen.

Automatic Control

Each evaporator is controlled from a temperature/humidity sensor located in the air return side of each unit. A temperature set point for each unit is entered at the Graphics Interface Computer. The system controller will add one (1) and subtract one (1) from the set point that is entered to give a two (2) degree dead band. The reheat solenoid operates from a humidity setpoint with offset. If the room humidity becomes greater than setpoint the reheat will come on until it reaches the offset. There are four (4) solenoid valves for each evaporator, individually controlled depending on the mode of the evaporators as described below. The units fan starters are controlled together with one (1) output from the controller at the Engine Room MCC.

A Satisfied temperature condition is true when the return air temperature for an evaporator is below the two (2) degree dead band. The units will remain Satisfied until the temperature has risen above the two (2) degree dead band.

Controller Action for Satisfied Units:

- Fans On or may cycle if Fan Cycling is enabled on Graphics Interface Computer.
- Liquid Valve Closed (de-energized).
- Suction Valve Open (energized).
- Hot Gas Valve Closed (de-energized).
- Re-Heat Valve, controlled by Humidity setpoint.

A Call for Cooling condition is true when the return air temperature is above the two (2) degree dead band. The units will continue to cool until the temperature falls below the two (2) degree dead band. The two-degree dead band is plus and minus one from the setpoint.

Controller Action for Units Cooling:

- Fans On, will override Fan Cycling.
- Liquid Valve Open (energized).
- Suction Valve Open (energized).
- Hot Gas Valve Closed (de-energized).
- Re-Heat Valve Open (energized).

Defrost Control

The Evaporators have Hot Gas to defrost the coils. The following is the defrost sequence and controller action which each of the units will go through when defrost is initiated:

Pump Out Cycle:

Pumpout will remove any existing ammonia from the coil and is typically set from 5 minutes to 30 minutes

- Fans On
- Liquid Valve Closed (de-energized).
- Suction Valve Open (energized).
- Hot Gas Valve Closed (de-energized)
- Re-Heat Valve Open (energized).

Hot Gas Cycle:

Removes any frost/ice that has formed on the coil and is typically set from 10 minutes to 30 minutes.

- Fans Off
- Liquid Valve Closed (de-energized).
- Suction Valve Closed (de-energized).
- Hot Gas Valve Open (energized)
- Re-Heat Valve Open (energized).

Equalize/Bleed Cycle:

Brings the coil back down from Discharge (hot gas) pressure to suction pressure and is typically set from 3 minutes to 10 minutes.

- Fans Off
- Liquid Valve Closed (de-energized).
- Suction Valve Closed (de-energized).
- Hot Gas Valve Closed (de-energized)
- Re-Heat Valve Open (energized).

Fan Delay Cycle Mode:

Chills the coil back down and freezes any water droplets on the coil before turning the fans back on. Typically set from 3 minutes to 5 minutes.

- Fans Off
- Liquid Valve Open (energized).
- Suction Valve Open (energized).
- Hot Gas Valve Closed (de-energized)
- Re-Heat Valve Open (energized).

Each portion of the defrost cycle can be adjusted from the Graphics Interface and is found on each unit settings screen shown below.

AH-23 SETTINGS

Temp StPnt: 45
High Alarm: 55
Low Alarm: 35
Humidity StPnt: 80

HEAT MODE SETTINGS

Heat Offset: 3
Disabled

Force Defrost
Unit Enabled
Exit Settings

FAN CYCLE SETTINGS
Disabled
(0 entered for Fan On Min. will keep fans off when air unit is satisfied and fan cycling is enabled)

TEMPERATURE SETBACK SETTINGS

	Sch 1	Sch 2	Sch 3	Sch 4	Sch 5	Sch 6	Sch 7
Temp:	0	35	0	0	0	0	47
Day In:	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled
Time In:	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled
Day Out:	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled
Time Out:	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled

Disabled

DEFROST SETTINGS (MINUTES)

Demand
Pump Out: 10
Hot Gas: 8
Bleed Down: 3
Fan Delay: 3

8 hrs.
Hours Between Defrost

The Evaporator Settings Screen

Defrost Groups

The plant has two(2) defrost groups with only one air unit in each group allowed to be in defrost. Groups are shown below.

GROUP 1 AU-2A, AU-2B, AU3A, AU-4A, AU4B, AU17A and AU-17B

GROUP 2 AU-9 through AU-16

Fan Cycling Button/Function

If Fan Cycling is not enabled the fans will remain on at all times. If Fan Cycling is enabled and the Evaporator is satisfied the fans will run for the amount of time entered into the On time and will cycle off for the amount of time entered into the Off time. Upon a call for cooling the fans will be turned on even if currently off from Fan Cycling. To keep fans off, whether or not fan cycling is enabled, when an air unit is satisfied a zero must be entered in the fan cycle on time.

Scheduled Defrost Button/Function

The Scheduled Defrost function allows a seven day defrost schedule to be entered at the Graphics Interface Computer. In this schedule an evaporator can be scheduled to defrost up to four (4) times daily every day of the week (Military time). The Defrost Type button is used to choose between a scheduled defrost or a demand type defrost.

Demand Defrost Button/Function

If the Demand Defrost is enabled the controller will accumulate the amount of time which has elapsed in the cooling cycle (Liquid Valve On Time). When this accumulated time reaches the Demand Interval Set Point, the unit will begin the defrost cycle. The Demand Defrost accumulated time will be reset upon entering the defrost cycle.

Force Defrost Button/Function

The Force Defrost function will allow for an Evaporator to be put into Defrost at any time.

Note: Only one evaporator can go into defrost in a zone at a time if more than one valve group services that zone.

Temperature Setback Button/Function

The Temperature Setback function will enable the Evaporators in a certain area to be set to a different temperature control point at a specific time. The Setback Schedule is a seven (7) day schedule which will allow the operator to set the time to Start and Stop the Setback for each day. If this function is not selected the evaporator will continue to operate at its original set point around the clock. The Maximum length for a schedule per day is 23Hrs 59Min.

Re-Heat Mode Button/Function

Re-Heat mode allows the operator to enable/disable the re-heat solenoid with a button. While enabled a setpoint value will turn the re-heat off if the temperature gets "x" degrees above the temperature set point. above

Unit Enable/Disable Button

When an Air Unit is Disabled it will go through a 5-minute Pumpout then all solenoids are De-Energized and fans turned off until its Enabled again. The toggle switches in control panels will override this function

Time Delay After Power Failure

Upon a power failure the Evaporators will delay five (5) seconds between each evaporator before restarting when power has been returned.

Manual Control

Switch Banks located in the Existing Refrigeration Control Panel enable manual control of each air units valves and fans. These switches must be placed in appropriate position for the desired refrigeration cycle (cooling/defrost). This feature should only be used if the PLC fails or some other fault condition causes the operator to need to override all automatic controls.

Alarms

Operator adjustable High Temperature Alarm set point.

Operator adjustable Low Temperature Alarm set point.

Fan Fail Start Alarm, fan told to start and no auxiliary conformation back from the starter.

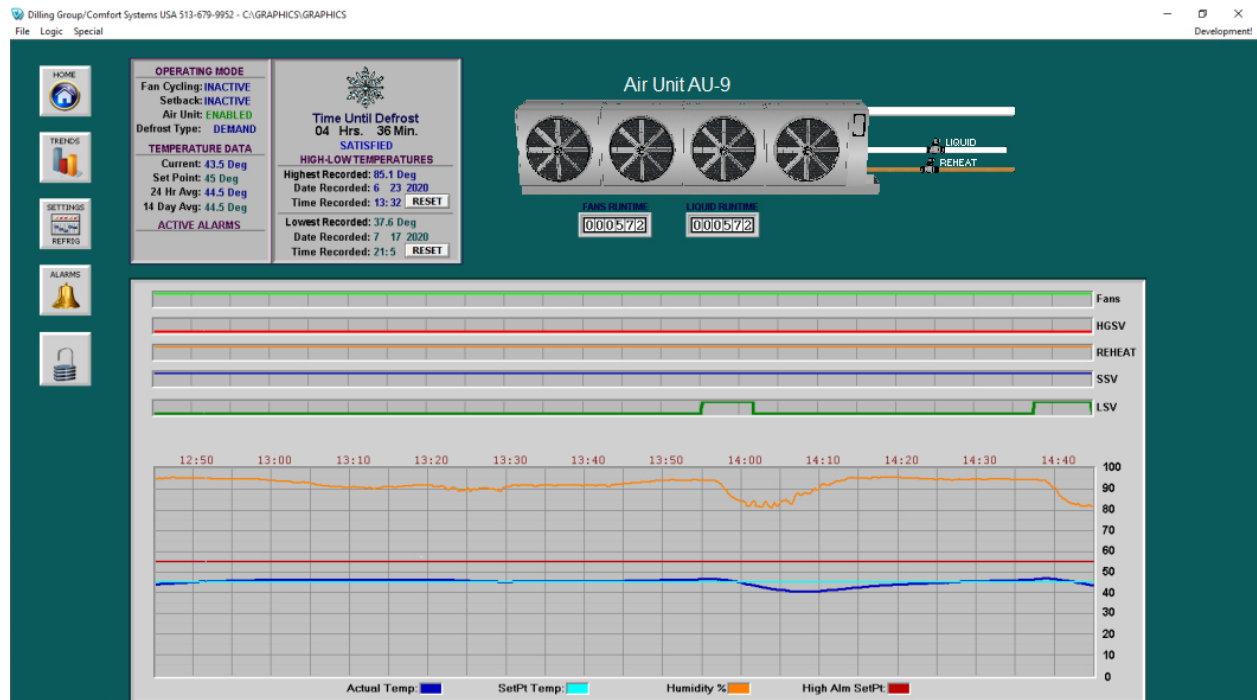
Out of Range Alarms

If a Temperature Sensors value to the PLC becomes too high or low (indicating bad sensor) the graphics will generate a Sensor Overrange or Under range alarm that will remain active until the problem is resolved.

2.1 AU-9 Through AU-16, Evaporative Air Units Control

Detail Screen

From the plant screen each air units Detail Screen can be selected by moving the mouse over the desired air unit and selecting with mouse click. From this screen you can see detailed information about an air unit including, 2-hour real time trend, Valve status, Fan status, Runtime meters for fans and for unit cooling. Operating Mode information, Temperature Data including 24 hour & 14 day running average, Time until next defrost, High/Low Temperature, time and date of highest and lowest values recorded. Runtime hours and High/Low temperature data can be reset once logged in. You also access the units setting and history trend from the buttons on the left.



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Automatic Control

Each evaporator is controlled from a temperature/humidity sensor located in the air return side of each unit. A temperature set point for each unit is entered at the Graphics Interface Computer. The system controller will add one (1) and subtract one (1) from the set point that is entered to give a two (2) degree dead band. The reheat solenoid operates from a humidity setpoint with offset. If the room humidity becomes greater than setpoint the reheat will come on until it reaches the offset. There are four (4) solenoid valves for each evaporator, individually controlled depending on the mode of the evaporators as described below. The units fan starters are controlled together with one (1) output from the controller at the Engine Room MCC.

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Controller Action for Satisfied Units:

- Fans On or may cycle if Fan Cycling is enabled on Graphics Interface Computer.
- Liquid Valve Closed (de-energized).
- Re-Heat Valve, controlled by Humidity setpoint.

A Call for Cooling condition is true when the return air temperature is above the two (2) degree dead band. The units will continue to cool until the temperature falls below the two (2) degree dead band. The two-degree dead band is plus and minus one from the setpoint.

Controller Action for Units Cooling:

- Fans On, will override Fan Cycling.
- Liquid Valve Open (energized).
- Re-Heat Valve Open (energized).

Defrost Control

The Evaporators have Hot Gas to defrost the coils. The following is the defrost sequence and controller action which each of the units will go through when defrost is initiated:

Pump Out Cycle:

Pumpout is used as a Air Defrost with this valve group type and is typically set from 5 minutes to 30 minutes

- Fans On
- Liquid Valve Closed (de-energized).
- Re-Heat Valve Open (energized).

Hot Gas Cycle:

There is no Hot Gas solenoid and set this to 1 minute.

- Fans Off
- Liquid Valve Closed (de-energized).
- Re-Heat Valve Open (energized).

Equalize/Bleed Cycle:

Air Defrost and set this to 1 minute

- Fans Off
- Liquid Valve Closed (de-energized).
- Re-Heat Valve Open (energized).

Fan Delay Cycle Mode:

Air Defrost and set this to 1 minute.

- Fans Off
- Liquid Valve Open (energized).
- Re-Heat Valve Open (energized).

Each portion of the defrost cycle can be adjusted from the Graphics Interface and is found on each unit settings screen shown below.

AH-23 SETTINGS

Temp StPnt: 45
High Alarm: 55
Low Alarm: 35
Humidity StPnt: 80

HEAT MODE SETTINGS

Heat Offset: 3
Disabled

FAN CYCLE SETTINGS

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TEMPERATURE SETBACK SETTINGS

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Temp:	0	35	0	0	0	0	47
Day In:	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled
Time In:	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled
Day Out:	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled
Time Out:	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled	Disabled

Disabled

DEFROST SETTINGS (MINUTES)

Demand Pump Out: 10 Hot Gas: 8
Bleed Down: 3 Fan Delay: 3

8 hrs.
Hours Between Defrost

Force Defrost
Unit Enabled
Exit Settings

The Evaporator Settings Screen

Defrost Groups

The plant has two(2) defrost groups with only one air unit in each group allowed to be in defrost. Groups are shown below.

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GROUP 2 AU-9 through AU-16

Fan Cycling Button/Function

If Fan Cycling is not enabled the fans will remain on at all times. If Fan Cycling is enabled and the Evaporator is satisfied the fans will run for the amount of time entered into the On time and will cycle off for the amount of time entered into the Off time. Upon a call for cooling the fans will be turned on even if currently off from Fan Cycling. To keep fans off, whether or not fan cycling is enabled, when an air unit is satisfied a zero must be entered in the fan cycle on time.

Scheduled Defrost Button/Function

The Scheduled Defrost function allows a seven day defrost schedule to be entered at the Graphics Interface Computer. In this schedule an evaporator can be scheduled to defrost up to four (4) times daily every day of the week (Military time). The Defrost Type button is used to choose between a scheduled defrost or a demand type defrost.

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The Temperature Setback function will enable the Evaporators in a certain area to be set to a different temperature control point at a specific time. The Setback Schedule is a seven (7) day schedule which will allow the operator to set the time to Start and Stop the Setback for each day. If this function is not selected the evaporator will continue to operate at its original set point around the clock. The Maximum length for a schedule per day is 23Hrs 59Min.

Re-Heat Mode Button/Function

Re-Heat mode allows the operator to enable/disable the re-heat solenoid with a button. While enabled a setpoint value will turn the re-heat off if the temperature gets "x" degrees above the temperature set point. above

Unit Enable/Disable Button

When an Air Unit is Disabled it will go through a 5-minute Pumpout then all solenoids are De-Energized and fans turned off until its Enabled again. The toggle switches in control panels will override this function

Time Delay After Power Failure

Upon a power failure the Evaporators will delay five (5) seconds between each evaporator before restarting when power has been returned.

Manual Control

Switch Banks located in the Existing Refrigeration Control Panel enable manual control of each air units valves and fans. These switches must be placed in appropriate position for the desired refrigeration cycle (cooling/defrost). This feature should only be used if the PLC fails or some other fault condition causes the operator to need to override all automatic controls.

Alarms

Operator adjustable High Temperature Alarm set point.

Operator adjustable Low Temperature Alarm set point.

Fan Fail Start Alarm, fan told to start and no auxiliary conformation back from the starter.

Out of Range Alarms

If a Temperature Sensors value to the PLC becomes too high or low (indicating bad sensor) the graphics will generate a Sensor Overrange or Under range alarm that will remain active until the problem is resolved.

3.0 Alarms/Events and Alarm Configuration

Each screen has an Alarm button. A current alarm is indicated by a flashing and ringing yellow bell on the Alarm button. A typical installation will also have a horn and beacon near the engine the alarm button. The alarm screen displays the date and time the alarm occurred, room to alert personnel in the area. Current alarms (example below) can be viewed by selecting

[illegible]

the operator that was logged on at the time of the alarm, and a description of the alarm. Once the condition is no longer in alarm, the entry will remove itself from this screen. The alarm can be acknowledged by selecting the "Alarm Ack" button. To acknowledge the Alarm, the operator must first "Logon" to the system using the appropriate password.

To view a history of alarms and events select the “Alarm History” button. The “Alarm History” screen (example below) shows a history of the alarms, when they occurred, when they returned (no longer an alarm condition), and when they were acknowledged. Events such as changing critical setpoints, forcing a defrost, Logon/Logoff are also recorded on this screen.

Date	Time	Name	Operator	Comment
06/12/15	11:52:03 AM	SuccessLevel	MONMOUTH-Pc/ecs	SuccessLevel
06/12/15	11:52:03 AM	OperatorName	MONMOUTH-Pc/ecs	OperatorName
06/12/15	11:52:03 AM	SOperator	MONMOUTH-Pc/None	SOperator
06/12/15	11:46:56 AM	SnackcityTimeout	MONMOUTH-Pc/None	SnackcityTimeout
06/12/15	11:36:30 AM	SnackcityTimeout	MONMOUTH-Pc/None	SnackcityTimeout
06/12/15	11:21:30 AM	SnackcityTimeout	MONMOUTH-Pc/None	SnackcityTimeout
06/12/15	11:21:10 AM	SnackcityTimeout	MONMOUTH-Pc/None	SnackcityTimeout
06/12/15	10:37:41 AM	VB314_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-314 Hh Temperature Alarm
06/12/15	10:36:01 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:35:16 AM	SnackcityTimeout	MONMOUTH-Pc/None	SnackcityTimeout
06/12/15	10:34:36 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:34:36 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:31:22 AM	SnackcityTimeout	MONMOUTH-Pc/None	SnackcityTimeout
06/12/15	10:30:16 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:30:16 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:25:06 AM	VB303_BlastUnload	MONMOUTH-Pc/None	VB-303 Blast Unload Button Pressed
06/12/15	10:20:23 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:20:23 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:17:51 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:17:51 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:16:22 AM	SnackcityTimeout	MONMOUTH-Pc/None	SnackcityTimeout
06/12/15	10:06:55 AM	SnackcityTimeout	MONMOUTH-Pc/None	SnackcityTimeout
06/12/15	10:05:42 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:00:42 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:07:38 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:07:38 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:04:22 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:04:22 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:03:20 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:03:20 AM	VB313_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-313 Hh Temperature Alarm
06/12/15	10:02:57 AM	VB314_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-314 Hh Temperature Alarm
06/12/15	10:02:57 AM	VB314_Hh_Temp_Alarm	MONMOUTH-Pc/None	VB-314 Hh Temperature Alarm

Update Successful

Default Query

UnAck Alarm

Ack Alarm

Alarm Return

Event

Alarm Ack

Exit

Alarm Summary

Alarm History Screen

In addition to displaying this information on the screen, all alarms and events are logged to a file on the computer that can be saved for a permanent alarm/event history.

Graphics Alarms Configuration

Each alarm in the graphics system can be configured to energize the Alarm Horn, energize the Alarm Beacon, activate an auto dialer if desired or disabled if equipment is being repaired. The following screens show example alarm configurations. In order for an alarm to be configured, the operator must first “Logon” to the system and has an adequate access level.

Alarm Configuration Screen 3

Evaporators			
VB-201 High Temp	☒ Light	☐ Horn	☐ Dial Out
VB-201 Low Temp	☒ Light	☐ Horn	☐ Dial Out
VB-201 Fan Failed Start	☒ Light	☐ Horn	☐ Dial Out
VB-202 High Temp	☒ Light	☐ Horn	☐ Dial Out
VB-202 Low Temp	☒ Light	☐ Horn	☐ Dial Out
VB-202 Fan Failed Start	☒ Light	☐ Horn	☐ Dial Out
VB-203 High Temp	☒ Light	☐ Horn	☐ Dial Out
VB-203 Low Temp	☒ Light	☐ Horn	☐ Dial Out
VB-203 Fan Failed Start	☒ Light	☐ Horn	☐ Dial Out
Future	☐ Light	☐ Horn	☐ Dial Out

Evaporators			
VB-204 High Temp	☒ Light	☐ Horn	☐ Dial Out
VB-204 Low Temp	☒ Light	☐ Horn	☐ Dial Out
VB-204 Fan Failed Start	☒ Light	☐ Horn	☐ Dial Out
VB-205 High Temp	☒ Light	☐ Horn	☐ Dial Out
VB-205 Low Temp	☒ Light	☐ Horn	☐ Dial Out
VB-205 Fan Failed Start	☒ Light	☐ Horn	☐ Dial Out
VB-206 High Temp	☒ Light	☐ Horn	☐ Dial Out
VB-206 Low Temp	☒ Light	☐ Horn	☐ Dial Out
VB-206 Fan Failed Start	☒ Light	☐ Horn	☐ Dial Out
Future	☐ Light	☐ Horn	☐ Dial Out

Evaporators			
VB-208 High Temp	☒ Light	☐ Horn	☐ Dial Out
VB-208 Low Temp	☒ Light	☐ Horn	☐ Dial Out
VB-208 Fan Failed Start	☒ Light	☐ Horn	☐ Dial Out
Future	☐ Light	☐ Horn	☐ Dial Out
Future	☐ Light	☐ Horn	☐ Dial Out
Future	☐ Light	☐ Horn	☐ Dial Out
Future	☐ Light	☐ Horn	☐ Dial Out
Future	☐ Light	☐ Horn	☐ Dial Out
Future	☐ Light	☐ Horn	☐ Dial Out

CONFIG 1

CONFIG 2

CONFIG 3

CONFIG 4

CONFIG 5

CONFIG 6

CONFIG 7

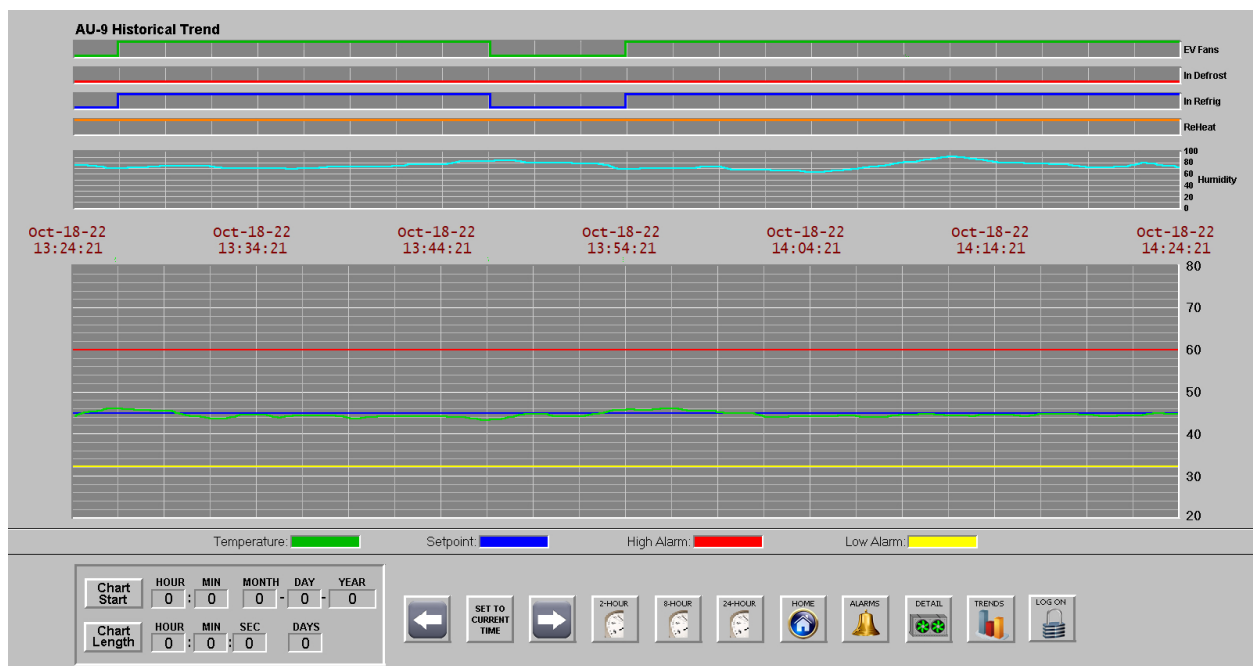
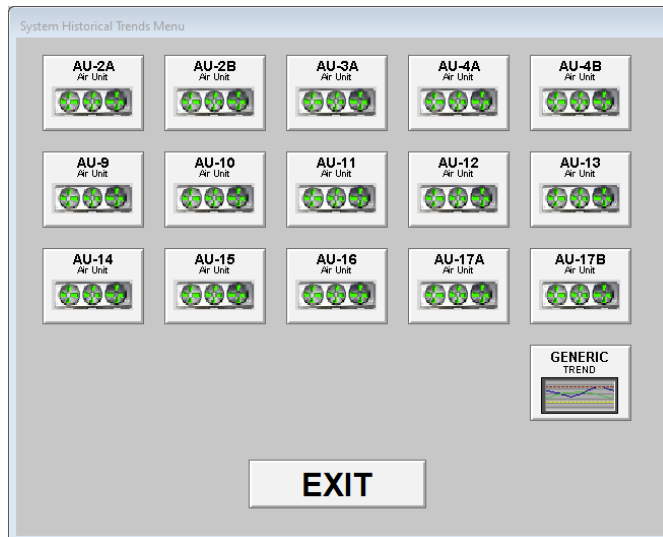
CONFIG 8

Date: 6/12/2015
Time: 12:08:06 PM

Alarm Configuration Screen

4.0 HISTORICAL TRENDS

Selecting the Trends button from Main or Condenser screen brings up the Trend Menu screen as shown below. If the operator selected the LSC button the screen below the Trends menu would be shown. From this screen an operator can look at historical information related to selected button. This information is saved for 10 years starting at the system startup/commission.



AU-9 History Trend screen



These buttons automatically set the screen length to 2, 8 or 24 hours.



With these buttons the trend can be set to current time or move the trend forward or back in time by the currently selected time frame. The current time is displayed on the right side of the trend.

Chart Start	HOUR	MIN	MONTH	DAY	YEAR
	0	0	0	0	0
Chart Length	HOUR	MIN	SEC	DAYS	
	0	0	0	0	

This area of the trend screen allow an operator to select any start time and trend length desired. The Start Chart and Chart Length must be selected to update the trend before selected information will be displayed

5.0 Parts List System Control Panel

<i>Item No.</i>	<i>Qty.</i>	<i>Description</i>	<i>Part Number</i>	<i>Manufacturer</i>
1	1	4 Slot ControlLogix Chassis	1756-A4	Allen Bradley
2	1	EtherNet/IP communication module, 10/100M twisted pair, 64 TCP connections	1756-ENBT	Allen Bradley
3	2	Slot Filler module for standard ControlLogix chassis	1756-N2	Allen Bradley
4	1	5094 Ethernet Adapter 16 Modules RJ45	5094-AEN2TR	Allen Bradley
5	2	5094 Digital 16 Input 120 VAC	5094-IA16	Allen Bradley
6	2	5094 Analog 8 input	5094-IF8	Allen Bradley
7	7	5094 Mounting Base	5094-MB	Allen Bradley
8	3	5094 Digital 16 Output 120/240 VAC	5094-OA16	Allen Bradley
9	5	5094 RTB AC spring	5094-RTB3ACS	Allen Bradley
10	2	5094 RTB Spring	5094-RTB3S	Allen Bradley
11	2	5094 Shield Bar Accessory	5094-STB	Allen Bradley
12	1	UL 489, Miniature Circuit Breaker, 1 Pole, C Curve, 40 Amp	1489-M1C400	Allen-Bradley
13	10	UL 1077, Miniature Circuit Breaker, 1 Pole, C Curve, 2 Amp	1492-SPM1C020	Allen-Bradley
14	5	UL 1077, Miniature Circuit Breaker, 1 Pole, C Curve, 5 Amp	1492-SPM1C050	Allen-Bradley
15	2	UL 489, Miniature Circuit Breaker, 2 Pole, C Curve, 16 Amp	1492-SPM2C160	Allen-Bradley
16	1	UL 489, Miniature Circuit Breaker, 2 Pole, C Curve, 50 Amp	1492-SPM2C500	Allen-Bradley
17	1	Controller,ControlLogix,3 MB User Memory,USB Port,1GB Ethernet port,100 EtherNet/IP Devices,4 Character Alpha/Numeric Display	1756-L81E	Allen-Bradley
18	1	ControlLogix, 19-32 VDC Power Supply (10 Amp @ 5V)	1756-PB72	Allen-Bradley
19	1	Stratix 2000 unmanaged switch, 16 copper 10/100 ports	1783-US16T	Allen-Bradley
20	1	Stratix 2000 unmanaged switch, 6 copper 10/100 ports, 2 Multimode 100 meg. preinstalled fiber SFP module	1783-US6T2F	Allen-Bradley
21	16	Fuse, 2 Amp, Glass for (2002-1611)	LTFS-218002P	Little Fuse
22	6	Wire Duct Cover 2"	C2LG6-F	Panduit
23	30	Wire Duct Cover 3"	C3LG6-F	Panduit
24	24	Wire Duct Cover 4"	C4LG6-F	Panduit
25	6	Wire Duct, 2" x 4" p/ft	F2X4LG6	Panduit
26	30	Wire Duct, 3" x 4" p/ft	F3X4LG6	Panduit

27	24	Wire Duct, 4" x 4" p/ft	F4X4LG6	Panduit
28	1	Ground Block for 14-2 Awg Wire	LAMA2-14-QY	Panduit
29	2	TeSys D control relay - 5 NO - <= 690 V - 110 V AC standard coil	CAD50F7	Schneider
30	2	TeSys D - safety protective cover - for LC1 D09...65	LAD9ET1S	Schneider
31	1	15 Pole Ground Block	PK15GTA	Square D
32	20	TOPJOB*S feedthrough terminal block; rail mount; 4-conductor; 6.2 mm wide; red	2002-1403	WAGO
33	2	TOPJOB*S feedthrough terminal block; rail mount; 4-conductor; 5.2 mm wide; blue	2002-1404	WAGO
34	10	TOPJOB*S ground terminal block; rail mount; 4-conductor; 5.2 mm wide; green-yellow	2002-1407	WAGO
35	3	End and intermediate plate; 1 mm thick; orange	2002-1492	WAGO
36	16	TOPJOB*S fuse disconnect terminal block; with pivoting fuse holder for 5 x 20 mm fuse; rail mount; 2-conductor; 6.2 mm wide; gray	2002-1611	WAGO
37	1	TOPJOB*S jumper; for 2002 series; insulated; 10-way; light-gray	2002-410	WAGO
38	1	TOPJOB*S end/intermediate plate; for 2002 series fuse blocks; 2.0 mm wide; orange	2002-992	WAGO
39	5	Steel carrier rail; 35 x 7.5 mm; 1 mm thick; 2 m long; slotted; according to EN 60715; "Hole width 25 mm	210-112	WAGO
40	22	Screwless end stop; 6 mm wide; for DIN-rail 35 x 15 and 35 x 7.5; gray	249-116	WAGO
41	2	Interface module; RJ-45; IDC technology; Cat. 6; in mounting carrier; with shield connection	289-195	WAGO
42	2	EPSITRON® ECO power supply; single-phase; output voltage 24 VDC; 20 A	787-734	WAGO
43	1	Diode redundancy module; output voltage 24 VDC; 40 A	787-785	WAGO
44	2	16 Port HOA Switch Board	STOCK	WAGO
45	2	Supply terminal, PUSH IN, 6 mm², 500 V, 41 A, Dark Beige (BLUE) AAP11 6 LO BL	1988130000	Weidmuller
46	20	Modular distribution terminals, PUSH IN, 1.5 mm², 500 V, 17.5 A, Dark Beige (RED) AAP11 1.5 LI RD	1988160000	Weidmuller
47	20	Modular distribution terminals, PUSH IN, 1.5 mm², 500 V, 17.5 A, Dark Beige (BLUE) AAP11 1.5 LI BL	1988170000	Weidmuller
48	4	A-series, End plate AEP AP11	1988320000	Weidmuller
49	2	Supply terminal, PUSH IN, 6 mm², 500 V, 41 A, Dark Beige (RED) AAP11 6 LO RD	1989780000	Weidmuller
50	6	A-series, End bracket AEB 35 SC/1	1991920000	Weidmuller